

Thoughts on Theory and Understanding

Theories are not permanent:

No fairer destiny [has] any physical theory, than that it should of itself point out the way to the introduction of a more comprehensive theory, in which it lives on as a limiting case.

Albert Einstein

We don't really care about understanding:

The only object of theory is to calculate results that can be compared with experiment. . . it is unnecessary that the whole course of the phenomenon be given.

Paul Dirac

Indeed, it may be hubris to think that we actually construct theories or understand:

The sciences do not try to explain. . . they mainly make models. The justification [of a model] is solely and precisely that it is expected to work.

John von Neumann